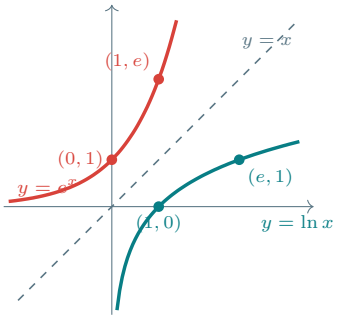
FUNCTIONS LADDER 5 OF 10

Logarithms

The logarithm undoes the exponential, turning products into sums and powers into multiples.

1 / THE RULE AND THE PICTURE

$\ln x$ is the inverse of e^x, so its graph is the mirror image in the line $y = x$



Swapping the coordinates of every point on $y = e^x$ gives the curve $y = \ln x$, defined for $x > 0$ only because e^y is always positive:

$$\ln(e^x) = x, \qquad e^{\ln x} = x, \qquad \ln 1 = 0, \quad \ln e = 1$$

The base-10 logarithm has $\log_{10} 10^x = x$, and $\log_a x$ is the power to which a must be raised to give x.

2 / THE LAWS / MULTIPLICATION BECOMES ADDITION

Each law is an exponential law read backwards

law	statement	example
product	$\log(ab) = \log a + \log b$	$\log_{10} 6 = 0.30103 + 0.47712 = 0.77815$
quotient	$\log(a/b) = \log a - \log b$	$\log_{10} 1.5 = 0.47712 - 0.30103$
power	$\log a^n = n \log a$	$\log_{10} 2^{10} = 3.0103$, and $2^{10} = 1024$
change of base	$\log_a x = \dfrac{\ln x}{\ln a}$	$\log_3 2 = \frac{0.6931}{1.0986} = 0.6309$

Since $e^{a+b} = e^a e^b$, taking logarithms gives the product law, and before calculators this turned long multiplications into additions read from tables, as on a slide rule.
Common values are $\log_{10} 2 = 0.30103$, $\log_{10} 3 = 0.47712$, $\log_{10} 5 = 0.69897$ and $\log_{10} 7 = 0.84510$.

3 / CALCULUS AND SMALL VALUES

The derivative of $\ln x$ is $1/x$, and $\ln(1 + x) \approx x$ for small x

Differentiating $e^{\ln x} = x$ gives $e^{\ln x} (\ln x)' = 1$, so $(\ln x)' = 1/x$, and hence $\int \frac{dx}{x} = \ln|x| + C$. For small x the series $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$ gives $\ln 1.05 \approx 0.0488$ and $\ln 1.07 = 0.0677$ against $0.07 - 0.0025 = 0.0676$, which is why a growth of 7% is about 0.07 in logarithmic terms.

4 / LOGARITHMIC SCALES

Equal steps on the scale are equal ratios

quantity	formula	meaning
decibels	$10 \log_{10}(P/P_0)$ for power	doubling the power adds 3.01 dB, and a factor 1000 adds 30 dB
amplitude in dB	$20 \log_{10}(A/A_0)$	doubling the amplitude adds 6.02 dB
Richter magnitude	$\log_{10}$ of amplitude	one unit is $10\times$ the amplitude and $10^{1.5} = 31.6\times$ the energy
pH	$-\log_{10}[\mathrm{H}^+]$	$[\mathrm{H}^+] = 10^{-7}$ gives 7; 3.2×10^{-4} gives 3.49

Such scales compress huge ranges of values into manageable numbers, and they match the way hearing and brightness respond to ratios.

5 / LOG-LOG PLOTS AND POWERS

A power law $y = ax^b$ becomes a straight line of slope b when both axes are logarithmic

For $y = 2x^{3/2}$ the values at $x = 1, 10, 100, 1000$ are $2,\ 63.2,\ 2000,\ 63\,246$, and $\log_{10} y = 0.301,\ 1.801,\ 3.301,\ 4.801$ rise by exactly 1.5 for each factor of ten in x.
The same trick solves $1.07^t = 2$ as $t = \ln 2 / \ln 1.07 = 10.24$, and counts digits: 2^{100} has $\lfloor 100 \log_{10} 2 \rfloor + 1 = 31$ digits.

ASIDE / SEMI-LOG

Plotting $\ln y$ against t turns an exponential $y = ae^{kt}$ into a straight line of slope k, the quickest test for exponential growth.

LINKS TO OTHER SHEETS

The inverse relation starts in *Exponential functions and the number e*; the logarithm of a complex number is in *Euler's formula and the complex exponential*.

Convention. $\ln$ is $\log_e$; $\log_{10}$ the common logarithm; $\lfloor\ \rfloor$ rounds down; P is power and A amplitude.